

Lyu Tang, Ph.D.

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🎓 EDUCATION AND TRAINING

University of Chinese Academy of Sciences <i>Ph.D. in Computer Application Technology</i>	2021 – 2025 <i>China</i>
Nanjing University <i>M.Sc. in Computer Technology</i>	2018 – 2021 <i>China</i>
Southwest Jiaotong University <i>B.Sc. in Computer Science and Technology</i>	2014 – 2018 <i>China</i>

👜 ACADEMIC POSITIONS

University of Alberta <i>Postdoctoral Fellow, Department of Electrical and Computer Engineering</i>	2025 – 2026 <i>Canada</i>
The University of British Columbia <i>Postdoctoral Fellow, Michael Smith Laboratories</i>	2026 – Present <i>Canada</i>

🔍 RESEARCH AREAS

Generative AI <i>Multimodal Large Language Models, AI for Science</i>	2024 – Present
High-level Vision <i>Image segmentation</i>	2019 – 2025
Low-level Vision <i>Video compression</i>	2021 – 2025

⚙️ RELEVANT CONTRIBUTIONS

Methodological Contributions

- Developed generative and foundation model based methods for multimodal learning, open world segmentation, and high dimensional visual understanding.
- Designed efficient and scalable learning frameworks spanning multimodal large language models, prompt based segmentation, and deep video compression.

Scholarly Contributions

- Published **35** papers in leading venues including CVPR, ICCV, ACM MM, AAAI, IJCAI, IJCV, TIP, TCSVT, TMM and TOMM.
- Accumulated more than **1250** citations on Google Scholar.
- Served as reviewer for ICML, ICLR, NeurIPS, AAAI, CVPR, ECCV, ICCV, ACM MM, TIP, IJCV, and TCSVT.

📖 SELECTED PUBLICATIONS (FIRST/CO-FIRST/CORRESPONDING AUTHORS)

Multimodal Large Language Models

- InfoTok: Regulating Information Flow for Capacity Constrained Shared Visual Tokenization in Unified MLLMs.
First author
Submitted to IEEE Transactions on Image Processing (TIP), 2026.
- Visual Text Compression as Measure Transport.
First author
Submitted to Conference on Neural Information Processing Systems (NeurIPS), 2026.

3. CoVP: Harnessing Multimodal Large Language Models for Zero Shot Camouflaged Object Detection.
First author
ACM Multimedia (ACM MM), 2024.
4. ASAM: Boosting Segment Anything Model with Adversarial Tuning.
Corresponding author
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024.
5. Towards Training Free Open World Segmentation via Image Prompting Foundation Models.
First author
International Journal of Computer Vision (IJCV), 2024.

High-level Vision

1. Boosting Vision State Space Model with Fractal Scanning.
Oral; Co first author; Corresponding author
AAAI Conference on Artificial Intelligence (AAAI), 2025.
2. Toward Stable Co Saliency Detection and Object Co Segmentation.
Corresponding author
IEEE Transactions on Image Processing (TIP), 2022.
3. Re Thinking the Relations in Co Saliency Detection.
First author
IEEE Transactions on Circuits and Systems for Video Technology (TCSVT), 2022.
4. Detecting Camouflaged Object in Frequency Domain.
Co first author; Corresponding author
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2022.
5. Disentangled High Quality Salient Object Detection.
First author
IEEE/CVF International Conference on Computer Vision (ICCV), 2021.
6. From Compositing to Real World: Transformer Based Natural Image Matting.
Corresponding author
IEEE Transactions on Circuits and Systems for Video Technology (TCSVT), 2024.

Low-level Vision

1. UVC: A Unified Deep Video Compression Framework.
First author
ACM Transactions on Multimedia Computing, Communications, and Applications (TOMM), 2024.
2. High Efficiency Deep Learning Based Video Compression.
First author
ACM Transactions on Multimedia Computing, Communications, and Applications (TOMM), 2024.
3. Scene Matters: Model Based Deep Video Compression.
First author
IEEE/CVF International Conference on Computer Vision (ICCV), 2023.
4. CANeRV: Content Adaptive Neural Representation for Video Compression.
First author
IEEE Transactions on Image Processing (TIP), 2026.

FUNDING AND AWARDS

National Scholarship	2016
Outstanding Graduate of Sichuan Province	2018